



St. Dominic's Catholic School for Girls



Tour de Maths 2022

Leg 1

Question Paper



5 Mark Questions

Question 1

M	A	T	H	S
	M	A	T	

The diagram shows a 5 by 5 table.

The top row contains the letters M, A, T, H and S.

The fourth row contains the letters M, A and T as shown.

The remaining squares must be filled with the letters M, A, T, H and S such that no row, column, or diagonal contains the same letter more than once.

Complete the table.

Question 2

How many zeros are there in 3001^2 , when written in full?

Question 3

The letters A to I represent numbers.

The numbers are added vertically and horizontally to give the numbers in the last row (17; P; Q) and the last column (17; 14; 15).

Calculate $P+Q$.

A	B	C	17
D	E	F	14
G	H	I	15
17	P	Q	

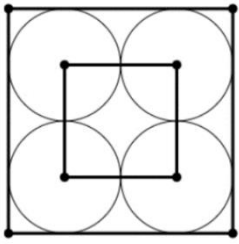
Question 4

The sequence 123456789123456789123... is continued until the sum of all the digits used is 460.

The last digit in the sequence is

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7
-

Question 5



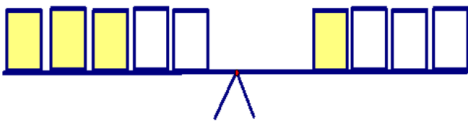
In the diagram, the congruent circles are tangent to the large square and each other as shown; and their centres are the vertices of the small square.

The area of the small square is 4.

Find the area of the large square.

Question 6

The diagram shows a perfectly balanced scale:



How many shaded rectangles should be placed on the right-hand side of the scale shown below to make it perfectly balanced?

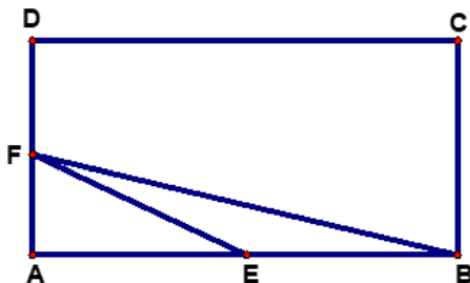


Question 7

I spent $\frac{3}{4}$ of my money, then $\frac{2}{5}$ of the rest and I still have R30 left.

How much did I spend?

Question 8



ABCD is a rectangle.

E and F are the midpoints of AB and AD in that order.

If the area of triangle BEF is 10cm^2 ,
calculate the area of rectangle ABCD, in square centimetres.

Question 9

I wrote down the integers 25, 26 27, ..., 208.

How many digits did I write down?

Question 10

In music a demisemiquaver is half of half of half a crotchet, and there are four crotchets in a semibreve.

How many demisemiquavers are there in a semibreve?

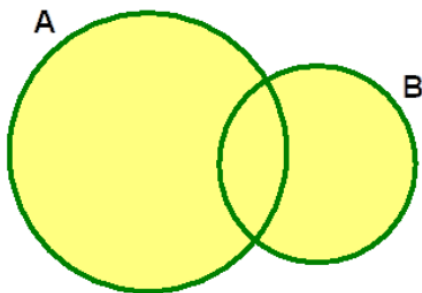
10 Mark Questions

Question 11

A palindromic number is one which reads the same when its digits are reversed, for example 15351.

What is the largest six-digit palindromic number which is exactly divisible by 15?

Question 12



The diagram shows two overlapping circles, A and B.

The area of the shaded region is $65m^2$ while the overlapping area is $15m^2$.

The ratio of the area of circle A to the area of circle B is 3:2.

By how many *square metres* is circle A larger than circle B?

Question 13

Two empty containers P and Q have the same volume.

Water flows into P at the rate of 4 litres per minute and into Q at the rate of 6 litres per minute.

After a certain time, container P can still take another 60 litres, but Q has overflowed by 10 litres.

What is the volume of each container?

Question 14

Sheba and Chante must dust all the ornaments (100 of them!) in their house every week.

If Sheba does it on her own, it takes her 1 hour and 40 minutes.

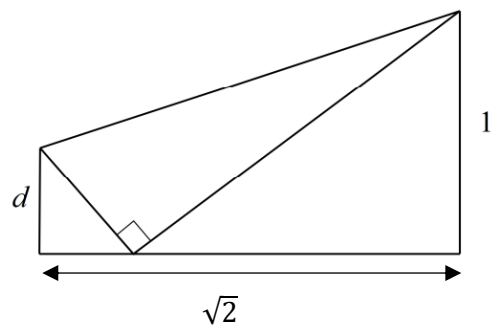
If Chante does it on her own, it takes her 3 hours and 20 minutes.

If they do it together how long will it take them?

Question 15

A rectangular sheet of paper with sides $\sqrt{2}$ and 1 has been folded as shown, so that one corner meets the opposite long edge.

The length d is ...



- (A) $\sqrt{2} - 1$ (B) $\sqrt{2} + 1$ (C) $2\sqrt{2} - 2$ (D) $\sqrt{3} - 1$ (E) $\sqrt{5} - 2$
-

20 Mark Questions

Question 16

Avery and Bryan have the following conversation:

Avery: I am thinking of two different one-digit numbers. Can you guess their sum?

Bryan: No. Can you give me a clue?

Avery: The last digit of their product is the same as your house number.

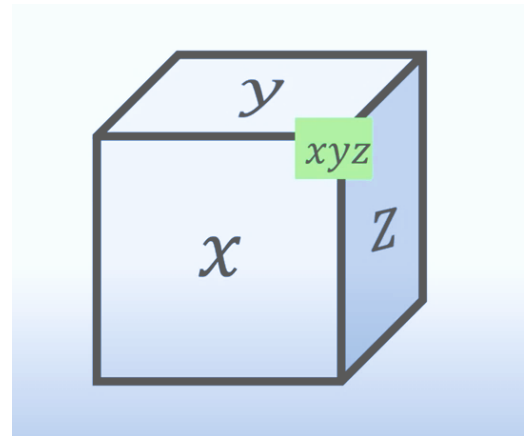
Bryan: Let me work it out ... Now I know!

What is the sum of the two numbers?

Question 17

A mathematician writes numbers on a cube. On each face she writes a positive integer (whole numbers greater than 0). On each vertex she writes the product of its three adjacent faces.

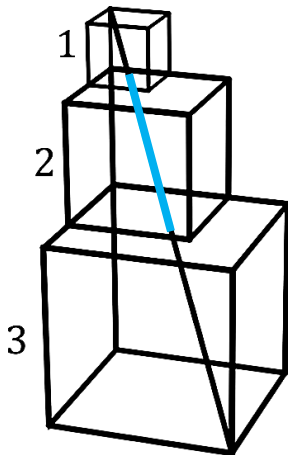
For example, in the upper right corner she would write the number that's equal to the product of x, y, z .



The mathematician does this for each of the vertices.

If the sum of all these numbers is 154, what is the sum of all the faces?

Question 18



A cube with edge length of 3 is stacked with a cube of edge length 2 and another cube with edge length of 1 as shown, where two of their faces are lined. Imagine drawing a line between opposite corners of the diagram.

What is the length of the line contained inside the middle cube?

Question 19

	Helen	Ivan
	64	104
	?	?

Helen and Ivan are carrying the same number of coins. Each coin is either a 20-cent coin or a 50-cent coin.

Helen has a total of 64 of the 20-cent coins and Ivan has a total of 104 of the 20-cent coins.

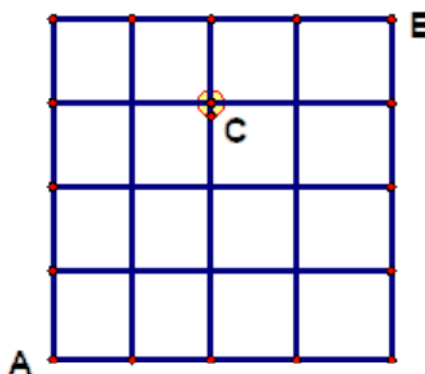
We don't know how many 50-cent coins either have.

Who has more money and by how much?

Question 20

The diagram shows a rectangular network of paths.

How many ways are there to move from point A to point B if we can only move up or right without moving through intersection C?



Total: 200

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